

Schaefer Technologies, Inc.

MODEL CP - 4200 CAPSULE POLISHER



Instruction Manual



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INSTRUCTION

CE INFORMATION, GENERAL RECOMMENDATIONS, SAFETY REMARKS and SPECIAL WARNINGS

INTRODUCTION

This Instruction Manual provides the user with information about how to use and maintain the machine correctly. It is compulsory to read this manual. It is our intention that the information contained in this manual will be a valid aid for your work. This information is based on data and on our best current knowledge. Carefully read this manual, including the recommendations and suggestions. No part of this manual may be copied or passed to third parties without the prior written consent of Schaefer Technologies, Inc.

We advise you to carefully read this manual, as it is essential to install, check and service the machine correctly in order to keep it in a perfectly efficient working order. The manual contains tables, designs and diagrams, which will help you to get to know the machine in detail. Please do not hesitate to contact us if you find that an explanation has been omitted or is unclear.

NAMEPLATE

For any servicing or overhaul operations not included in this manual or for any other technical problem which may arise, our Technical Personnel are available to provide you with information and to discuss any necessary action to take. When contacting us, please indicate the data on the machine's nameplate.

PRELIMINARY REMARKS

- The drawings and designs which show the system should be considered as reference and not necessarily faithful in every detail.
- The system dimensions and specifications contained in this manual are not binding and may be changed without warning.
- The designs and all of the other documents supplied as part of the system are the property of Schaefer Technologies, Inc.

The system is covered by a guarantee and a contract of purchase. During the period in which the machine is under guarantee, any repairs which have not been authorized by Schaefer Technologies, Inc. and are carried out on the same machine will automatically make the warranty void.

The user must take particular note of the safety warnings and information indicated by the following symbols and labels in this manual.



Mandatory.



Danger.



Lockout.



Prohibited.



Read manual.



Wear eye protection.



Wear safety gloves.

120 VOLTS

Electrical Voltage Specification. 120 Volts.

220 VOLTS

Electrical Voltage Specification. 220 Volts.



Electrical Current Specification. Single Phase.



Caution – Pinch point hazard. Keep hands clear.



Warning – Rotating parts inside. Keep hands clear. Lockout/tagout before servicing.



Warning – Hazardous Voltage.



Warning – Avoid injury. Do not operate with guards removed. Replace guards before operating machine. For the safety of the Personal. DO NOT modify or remove any of this equipment's guards, safety devices or warning labels.



Warning – Read and understand operator's manual before using this machine. Failure to follow operating instructions could result in injury or damage to the equipment.

GENERAL SAFETY REGULATIONS

- THESE SAFETY REGULATIONS HAVE BEEN DRAWN UP FOR YOUR OWN SAFETY. The strict observance of the safety regulations will reduce the risks of accidents for both you and or others.
- DO NOT attempt to move, install or operate the system without having first read and fully understood this manual. If in doubt, ask your supervisor.
- Make sure that all of the guards and protective covers are in place BEFORE starting the machine.
- NEVER leave tools, mechanical parts or other unsuitable material on or in the machine.
- If a failure should occur, push the stop button.
NEVER WORK ON THE MACHINE WHILE IT IS RUNNING.
- Pay particular attention even when the main switch is in the “OFF” position because the electrical power supply cables are still live.
- BEFORE restarting the production cycle after carrying out maintenance operations or repairs on the machine, make sure that all of the guards and protective covers are fitted correctly.
- Always be cautious, remember that your safety and that of fellow workers depend on you.
- When moving or lifting components of the machine, make sure that all the regulations for these operations are respected.

SPECIAL WARNINGS

- The personnel who operate this machine must know and scrupulously observe the general safety regulations. If these regulations are not observed, this may cause injury to workers and damage components of the machine.
- The maintenance operations must be carried out with the machine at a standstill. Lockout machine prior to servicing. Failure to follow operating instructions could result in personal injury or damage to the equipment.
- The user must make sure that all of the instructions in the manual are followed to the letter.
- Any tampering with the safety systems of the machine done for any reason, will be at the user’s own responsibility and risk.
- Do not ever try to bypass the safety systems or make them ineffective.
- For the machine smooth operation it is absolutely necessary to observe the speed, temperature and tension limits, along with all the indications given which the client must in any case respect.
- Furthermore, the environmental conditions must be taken into account for the installation of the machine.
- The national laws governing the use of this type of machine must also be observed.

WARNING

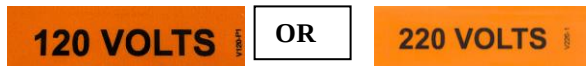
Schaefer Technologies, Inc. declines any responsibility for damage which may derive from the non-compliance with the above-mentioned warnings.

The declaration of conformity enclosed with this manual has been compiled by the manufacturer in compliance with the European Directive and defines the machine as “safe”.

The manufacturer’s declaration will become null and void if the machine is tampered with, modified, has its fixed and moving protective guards removed or is used for purposes other than those for which it was designed. The manufacturer disclaims all responsibility for any accidents or damage that may arise.

WARNING

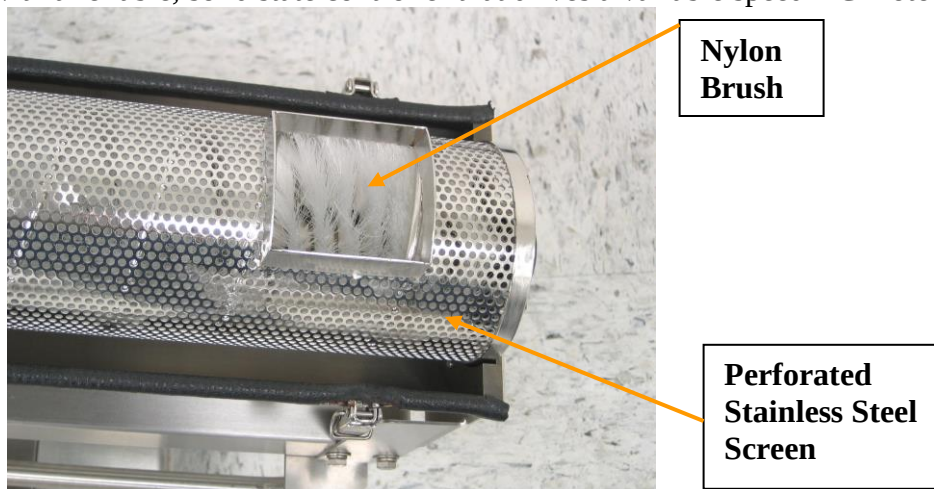
Read and understand operation manual prior to operating or servicing this machine. Failure to follow operating instructions could result in personal injury or damage to the equipment.



OVERVIEW

The Schaefer Technologies, Inc. Model CP-4200 Capsule Polisher is easy to operate, clean and maintain. The polisher is fast, clean and adaptable to batch or in-line processing. These factors make the CP-4200 the best buy among the capsule polishers available on the market today.

Connect to the appropriate power source for your specific machine (see the Power Wiring Diagram on page 16). As a standard, the CP-4200 requires only 120 volt, single phase, 60 Hz electrical service, but it can be supplied with other electrical configurations upon request. The machine is equipped with a reliable, solid state controller that drives a variable speed AC motor.



View Shown with Cover Removed

The CP-4200 polishes your capsules through the action of a soft nylon bristled brush that also conveys the capsules through the polisher. Dust and powder are conveyed with a vacuum assist through a perforated stainless steel screen. The two dust collecting nozzles may be connected either to house vacuum or to a portable vacuum available separately from Schaefer Technologies, Inc. All product contact parts, other than the nylon brush bristles, are of stainless steel.

With both vertical height and polishing chamber pitch adjustable, the CP-4200 can be used whether as an in-line or a batch polisher. The combination of pitch and height adjustment allows the CP-4200 to be used with the widest range of capsule filling machines: automatic and semi-automatic.

Although the pitch of the polishing chamber is adjustable, care should be taken so as not to make the pitch so steep as to allow the capsules to collect at the in-feed end of the chamber. This may lead to damage to the polisher and/or the product. Generally, the pitch should not exceed 30 degrees off the horizontal.

The CP-4200 is delivered completely assembled and ready to operate.

Unpacking, Set-up, and Operation

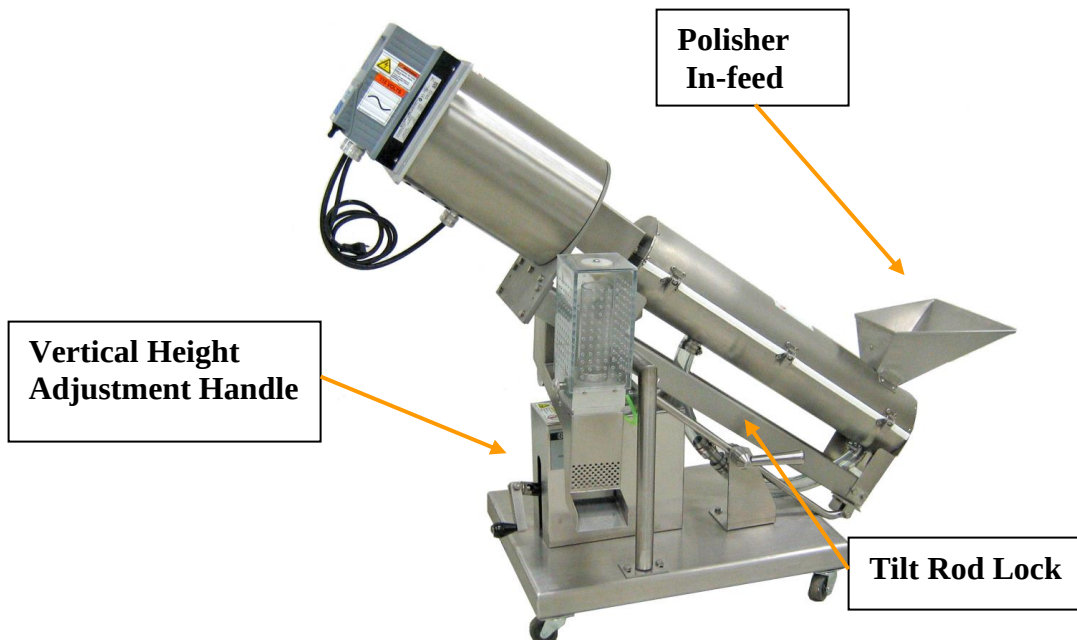
The Schaefer Technologies, Inc. CP-4200 Capsule Polisher is carefully packed so as to minimize transportation damage. After removal from the crate, carefully inspect the polisher for visible signs of damage.

Shipment contains:

- One - CP-4200 Capsule Polisher, completely assembled and operational.
- Two – Brush holding / drying cradles.
- One – Instruction Manual.

Be sure to remove the brush drying and holding cradles. When the brush is out of the polisher, these cradles are required to support the brush to prevent the bristles from taking a “set.” It is especially important that the cradles be used after washing to allow the brush to dry and retain its natural state.

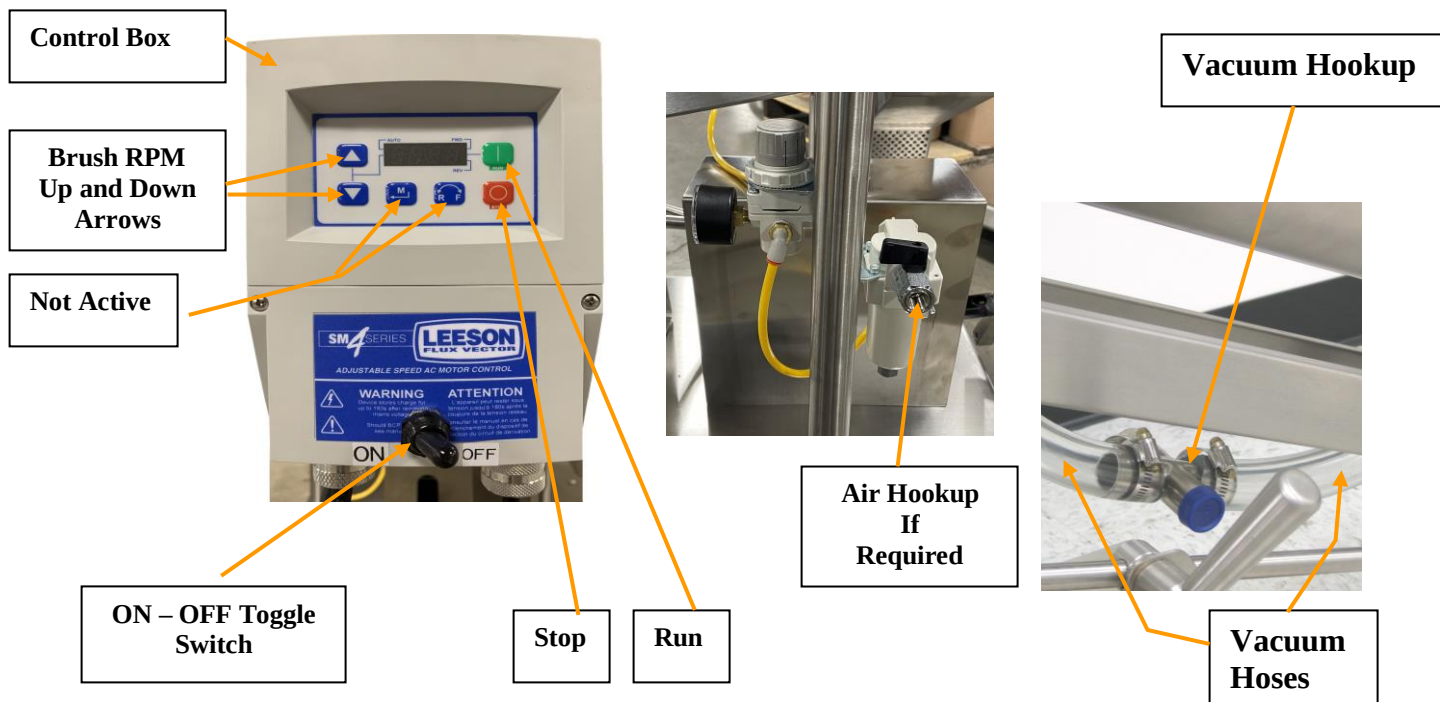
Set the machine on a level surface and set the pitch and height of the polisher to fit your application. The CP-4200 is designed for use either in batch polishing or as an in-line polisher. The height and pitch of the polisher can be adjusted to fit under a range of capsule filling machines: including semiautomatic fillers like the STI Models 10. When setting the CP-4200 for use with a Model 10, it is necessary to increase the vertical height slightly in order to fit the in-feed of the polisher under the out-feed of the capsule filler.



To adjust the height, turn the handle on the vertical height adjustment box. To adjust the pitch, loosen the tilt rod lock and set the pitch to the desired angle. Loosen the tilt rod lock carefully so as to best control the change in pitch and to prevent sudden large movements of the polishing chamber.

Unpacking, Set-up, and Operation (continued)

Connect the polisher to the appropriate power source for your machine (see the Power Wiring Diagram on page 15). If desired, attach vacuum lines to the polishing chamber exhaust and the extractor. The polisher performs best when vacuum is applied to remove dust and any item such as empty capsules or loose caps and bodies. If a suitable vacuum source is not available, Schaefer Technologies, Inc. offers a portable unit to provide the necessary vacuum source. Attach the air source for the Empty Capsule Eliminator and/or Air Flow Brush refer to those sections for the initial settings.



NOTE: When running the numbers will show the speed setting (in HZ). When not running the word "Stop" should be displayed. If for some reason the button "M" is pressed by mistake and "Stop" is NOT displayed, the "M" should be pressed again until "Stop" is displayed. (Machine will NOT start until "STOP" is in the display).

Move the ON/OFF switch to the "ON" position. Push "Run" on the control box and adjust the up or down arrows on the variable speed controller to obtain the optimum throughput speeds for your application. It is usually not necessary to set the rotational speed of the brush so that the rate of discharge of the capsules is greater than the input speed. Matching the input and output rates maximizes the ability of the CP-4200 to polish the capsules.

The pickup can be adjusted to maximize removal of powder/dust, not removed through the screen in the polishing chamber, while minimizing rejection of good capsules. The best way to monitor the setting is to put an air break, such as a glass or plastic bottle, in the vacuum line to collect the items.

DO NOT INSERT FINGERS OR OTHER OBJECTS INTO THE POLISHING CHAMBER WHILE THE POLISHER IS RUNNING.

Cleanup

When your production run is complete, the machine is easily disassembled for inspection and cleaning. If your changeover is between batches only, the following procedure is recommended:

1. Turn the speed control to the “Stop” position, move the ON/OFF switch to the “OFF” position and unplug the CP-4200. Remove the top cylinder half by opening the six (6) cam latches. Inspect and clean as necessary.
1. Remove the shaft coupling guard and loosen the allen set screws that hold the coupling on the brush shaft (the motor coupling does not have to be loosened). Disconnect the coupling and lift out the brush and polisher tube assembly. Slide the brush out of the polisher tube and inspect for strangers. Wash the brush if desired. The brush may be washed in any mild detergent or dish liquid. **Washing with solvents is not recommended.** If the brush is washed, dry it on the cradles provided to prevent distortion of the bristles.
2. Inspect the base cylinder half for strangers.
3. Disconnect the vacuum lines from the base cylinder half. Inspect the lines for strangers and reattach.
4. See page 17 for the empty capsule eliminator’s cleaning instructions.
5. See page 19 for the Air Flow Brush cleaning instructions.
6. To reinstall, slide a clean brush into the polisher tube assembly. Set the bearings into the blocks on each end of the base cylinder half. Make sure that the bearings are securely seated in the blocks. Reconnect the coupling making sure that the coupling is fully engaged and that the allen set screws are tight. Double-check the tightness of the set- screws in the couplings. Reattach the coupling cover.
7. Reattach the top cylinder half making sure that the bearings are fully seated in their blocks. Secure all the cam latches.
8. Inspect all the external surfaces of the polisher, including the base, for strangers: especially in the corners formed by the frame weldment and the end brackets.
9. Plug the CP-4200 back in the electrical source and you are ready to process the next batch of capsules.

Cleanup (continued)

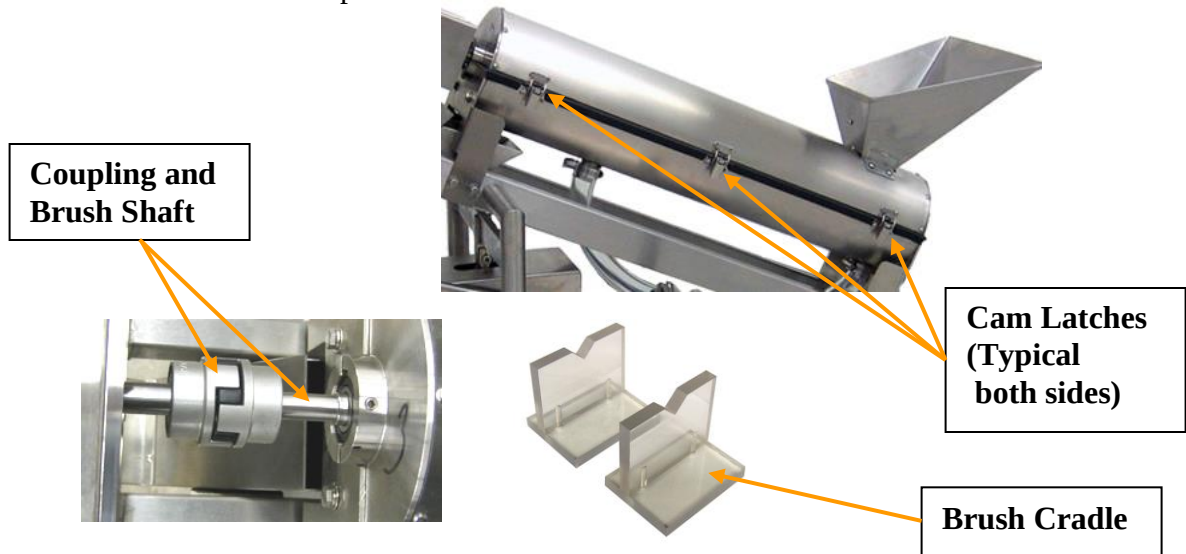
If your changeover is between products or different colors and/or strengths of the same product, the following procedure is recommended:

1. Move the ON/OFF switch to the “OFF” position. Turn the speed control to the “Stop” position and unplug the CP-4200. Remove the top cylinder half by opening the six (6) cam latches. Wash the top cylinder half.
2. Remove the shaft coupling guard and loosen the allen set screws that hold the coupling on the brush shaft (the motor coupling does not have to be loosened). Disconnect the coupling and lift out the brush and polisher tube assembly. Slide the brush out of the polisher tube and inspect for strangers. Wash the brush. The brush may be washed in any mild detergent or dish liquid. Washing with solvents is not recommended. Dry the brush on the cradles provided to prevent distortion of the bristles. Wash the polisher tube.
3. Inspect the base cylinder half for strangers.
4. Disconnect the vacuum lines from the base cylinder half and the dust extractor. Inspect the lines for strangers and reattach.
5. See page 17 for the Empty Capsule Eliminator cleaning instructions.
6. See page 19 for the Air Flow Brush cleaning instructions.
7. To reinstall, slide a clean brush into the polisher tube assembly. Set the bearings into the blocks on each end of the base cylinder half. Make sure that the bearings are securely seated in the blocks. Reconnect the coupling making sure that the coupling is fully engaged and that the allen set screws are tight. Double-check the tightness of the set- screws in the couplings. Reattach the coupling cover.
8. Reattach the top cylinder half making sure that the bearings are fully seated in their blocks. Secure all the cam latches.
9. Inspect all the external surfaces of the polisher, including the base, for strangers: especially in the corners formed by the frame weldment and the end brackets.
10. Plug the CP-4200 back in the electrical source and you are ready to process the next batch of capsules.

Maintenance and Troubleshooting

NOTE: The controller program is password protected. This allows the operator to access diagnostic parameters without being able to access the factory operational settings. Diagnostic parameters are listed in the AC Tech manual supplied with the machine. Please contact your Schaefer Technologies representative for any additional issues.

1. When setting up the CP-4200 for use with a capsule filler with a low discharge, such as the STI Models 10, it may be necessary to raise the polisher slightly so that the pitch can be adjusted to the filler's output chute.
2. The center of gravity of the polishing chamber and motor/controller is not precise. **Exercise caution in loosening the tilt rod lock to prevent sudden movements of the polishing chamber.** Loosening the rod lock slowly will also allow for easier setting of the pitch of the polishing chamber.
3. Brush maintenance and change out:
 - a. Remove the shaft coupling guard and loosen the allen set screws that hold the coupling on the brush shaft (the motor coupling does not have to be loosened). Disconnect the coupling and lift out the brush and polisher tube assembly. Slide the brush out of the polisher tube.

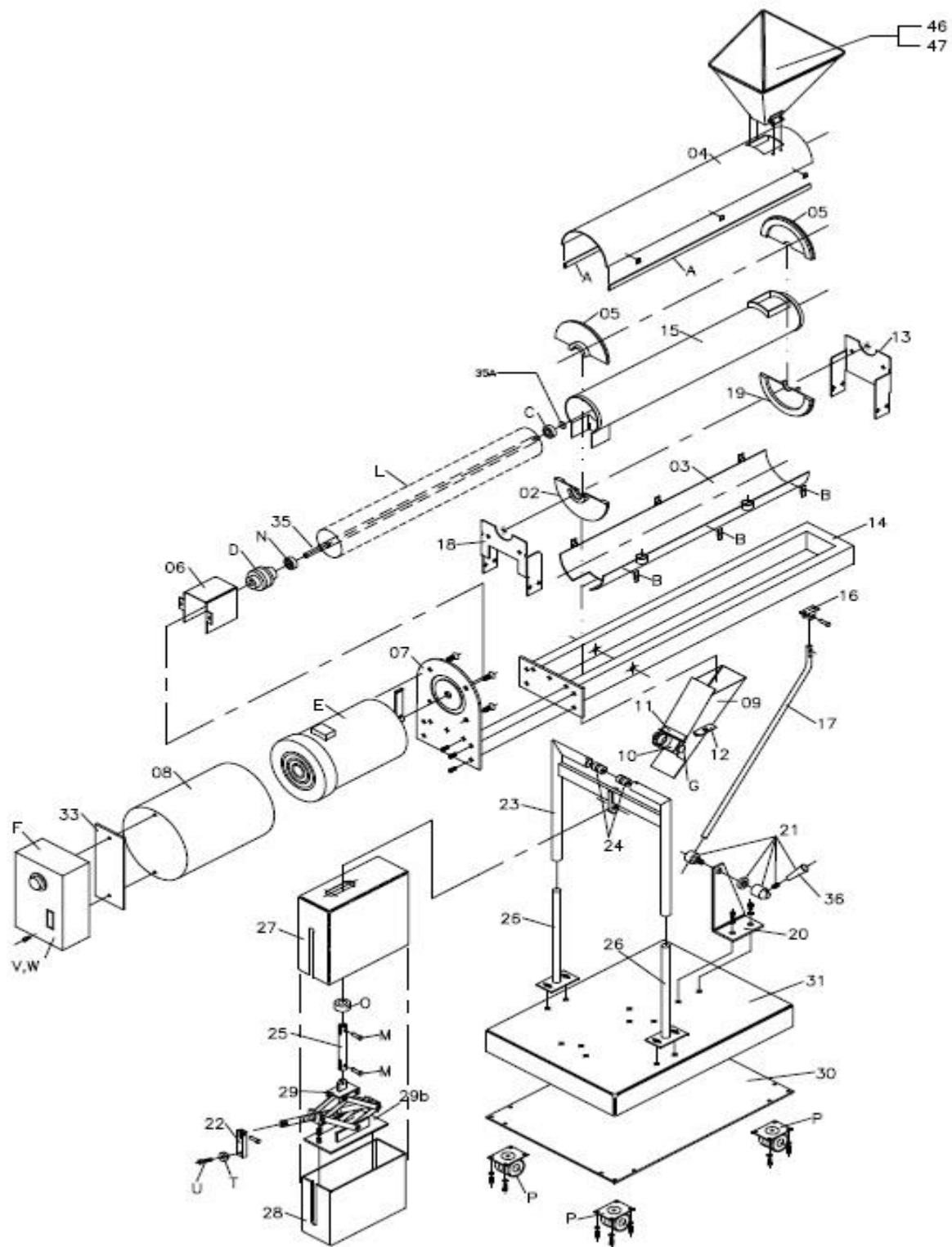


- b. The brush may be washed in any mild detergent or dish liquid. **Washing with solvents is not recommended.** If the brush is washed, dry it on the cradles provided to prevent distortion of the bristles. Air dry only.

Maintenance and Troubleshooting (continued)

- c. To reinstall, slide a clean brush into the polisher tube assembly. Set the bearings into blocks on each end of the base cylinder half. Make sure that the bearings are securely seated in the blocks. Reconnect the couplings making sure that the bearings are fully seated in their blocks. Secure all of the cam latches.
 - d. If you have more than one brush, be sure to store all spares in a cradle that prevents the bristles from bearing the full weight of the brush.
- 4. When using the CP-4200 as a batch polisher for larger capsules such as a double ought (00), the in-feed hopper may become clogged. If this occurs, gently swirl the capsules to encourage them to flow through the in-feed opening.
- 5. DO NOT INSERT FINGERS OR OTHER OBJECTS INTO THE POLISHING CHAMBER WHILE THE POLISHER IS RUNNING.**
- 6. If output from the polisher slows down, it is possible that:
 - a. The powder/dust pickup is set to close the out-feed chute; adjust as required.
 - b. Capsules are clogged in the in-feed hopper
 - c. The variable speed controller may be set at too low a setting to allow the polisher to keep up with the rate of capsule in-feed
 - d. The motor couplings may have been inadvertently left loose during reassembly; turn off the controller, unplug the CP-4200, and tighten the couplings as required; plug the polisher back in and adjust the speed controller to deliver the desired output.
- 7. It is normal for the motor housing to feel warmer to the touch at slower operating speeds than at higher ones. In any event, make sure that the vents on the bottom of the housing are clear.
- 8. If the bearings “chatter” or are otherwise noisy, make sure that they are properly seated in the two cylinder halves.
- 9. If the capsules exiting the CP-4200 are not sufficiently clean, make sure that the vacuum lines are attached and are not obstructed. Also, make sure that the rotational speed of the brush is not set too high compared to the rate of capsule in-feed. Reduced residence time in the polishing chamber will reduce the polisher’s performance.
- 10. If you have performance or operational questions contact Schaefer Technologies, Inc. at 317-546-4081.

Exploded View Drawing



V. Parts List use with Drawing on page 14

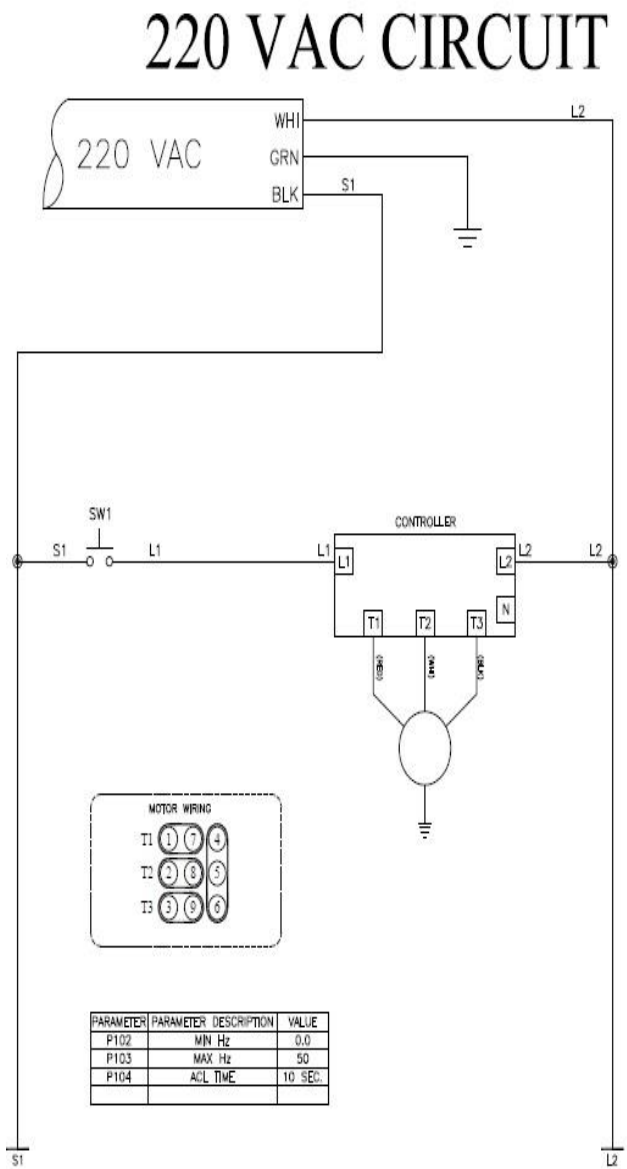
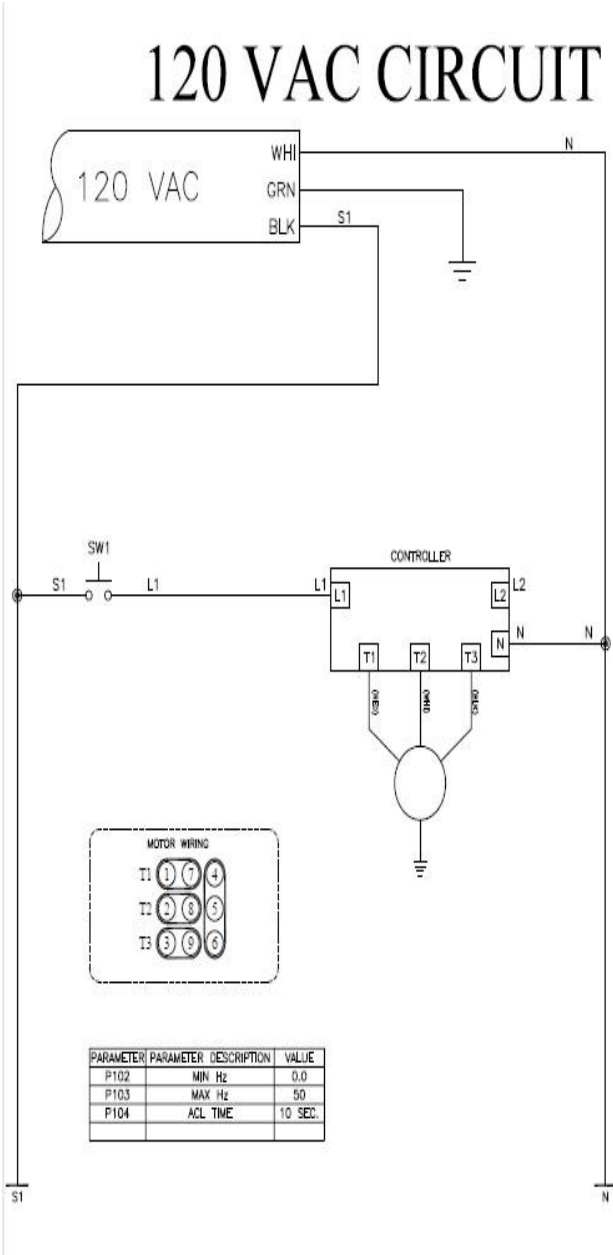
Item#	Description	Qty	Part #
02	Lower Tube End w/cut out	1	CP0002
03	Lower Tube Body	1	CP0003
04	Upper Tube Body	1	CP0004
05	Top Tube End Cap	2	CP0005
06	Coupling Guard	1	CP0006
07	Motor Mounting Plate	1	CP0007
08	Motor Cover	1	CP0008
09	Outlet Chute	1	CP0009
10	Plenum	1	CP0010
11	Plenum Bracket	1	CP0011
12	Mounting Bracket	2	CP0012
13	Closed End Bracket	1	CP0013
14	Main Frame Weldment	1	CP0014
15	Brush Tube Weldment	1	CP0015
16	Tilt Rod Bracket	1	CP0016
17	Tilt Rod	1	CP0017
18	Open End Bracket	1	CP0018
19	Tube End, Lower	1	CP0019
20	Base Bracket	1	CP0020
21	Rod Lock Assembly	1	CP0021
22	Jack Crank Arm	1	CP0022
23	Support Legs	1	CP0023
24	Pivot Hub	2	CP0024
25	Connecting Rod	1	CP0025
26	Lower Leg	2	CP0026
27	Jack Cover, Upper	1	CP0027
28	Jack Cover, Lower	1	CP0028
29	Jack Assembly	1	CP0029
30	Base Cover	1	CP0030
31	Base Weldment	1	CP0031
32	Brush Drying Bracket	2	CP0032
35	Brush Shaft	1	CP0035
35A	Bearing Retainer	1	CP0035A
36	Handle	1	CP0072
33	Controller MTG Plate	1	CP0084
46	Input Hopper	1	CP0096
47	Finger Guard, Hopper	1	CP0097

Item#	Description	Qty	Part#
A	U-Gasket 28"	2	CP0050
B	Cam Latch	6	CP0051
C	Bearing Lower	1	CP0100
D	Coupler (Motor Side) 53	1	CP0053
D	Coupler (Brush Side) 54	1	CP0054
D	Coupler (Spider) 55	1	CP0055
E	Motor	1	CP0056
G	Hand Knob	2	CP0058
*H	Tee Union	1	CP0059
*J	Vacuum Hose	26"	CP0060
*K	Hose Clamp	4	CF2797
L	Brush & Shaft Assembly	1	CP0063
M	3/8 X 1 Shoulder SCR.	2	CP0064
N	Bearing -Upper	1	CP0101
O	Shaft Collar	1	CP0066
P	Swivel Caster	4	CP0067
*Q	3/8" Cord Grip	3	CF2751
*R	1/2" Lock Nut	3	CF2807
*S	Elect. Cord (Power)	1	CP0081
T	Hand Knob	1	CP0070
U	3/8 X 1.5 SHLDR.SCR	1	CP0071
*V	Witch Seal & Boot Black	1	CP0079
*W	Toggle Switch SPST	1	CP0078
F	Motor Controller LEESON	1	CP0083
*	CE Tag	1	CP3022
*	Guard Label	1	SL0002
*	Rotating Parts Label		SL0003
*	Pinch Point Label		SL0014
*	Owner Manual Label		SL0005
*	Hazardous Voltage Label		SL0007
*	Single Phase Label		SL0013
*	120 Volt Label		SL0029
*	220 Volt Label		SL0016

Highlighted Items are recommended Spare Parts

NOTE: * = Not Shown in drawing

Electrical Diagram



Empty Capsule Eliminator

Warning: Do not over tighten Black Tip Nuts. Maximum torque is 24 in/lb.

Operation:

The capsule eliminator needs to run in the fully tilted (30 deg. Max) position to operate properly. The empty capsules removal is controlled by the speed of the brush and the air flow introduced into the unit. Depending on the size and weight of the capsules, the air pressure (normally 20psi +) and the speed of the brush (normally no less than 30Hz) will need to be adjusted.

The feed rate of the capsules being introduced into the capsule polisher hopper has a direct influence on the operation. If capsules are fed into the polisher too fast, a jam can occur at the entry into the empty capsule eliminator.

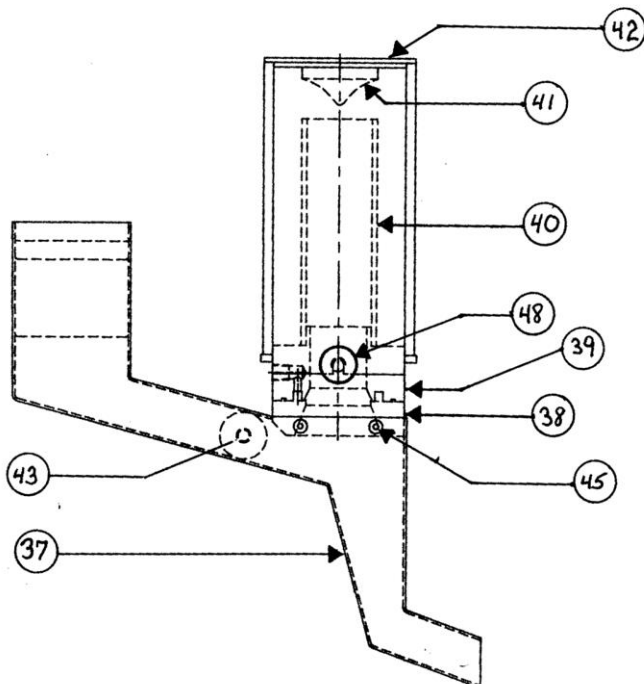
Use the following setting as a starting point:

Speed of brush - 35Hz

Air regulator - 22 p.s.i.

Cleaning (Disassembly): Note: Cleaning agent must be compatible with polycarbonate, aluminum, delrin and stainless steel.

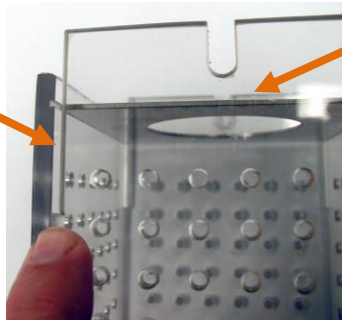
1. Remove the air line from the rear of the capsule eliminator.
2. Loosen the two knurled knobs (48) and remove the eliminator cover (42)
3. Remove the eliminator tube (40)
4. Remove four screws (45) from the eliminator base (38) and remove from the product chute (37).
5. Remove the bolt (43) attaching the product chute to the leg brace of the capsule polisher.



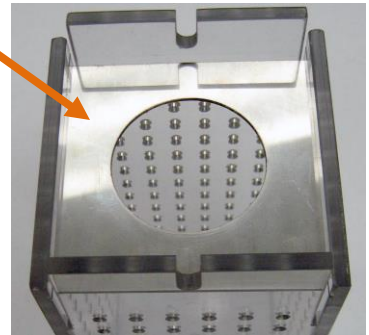
Item#	Description	Qty	Part #
37	Product Chute	1	CP0085
38	Eliminator Base	1	CP0086
39	Eliminator Manifold	1	CP0087
40	Eliminator Tube	1	CP0088
41	Deflector	1	CP0089
42	Eliminator Cover	1	CP0090
44	Inlet Cover	1	CP0092
45	Adapter Plate	1	CP0095

6. Disassemble the eliminator cover (42) per instructions below.

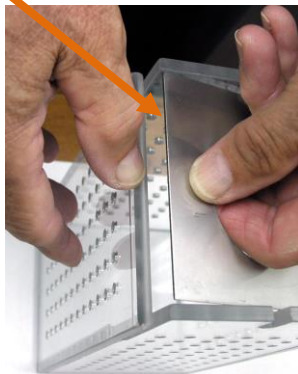
Det (42) – Deep Slot
(4) each. allows
flexing to enable the
removal and install
of Det (42D)
Stainless Plate



Det (42D)



Gripping cover and
plate as shown – flex
cover and remove
plate.



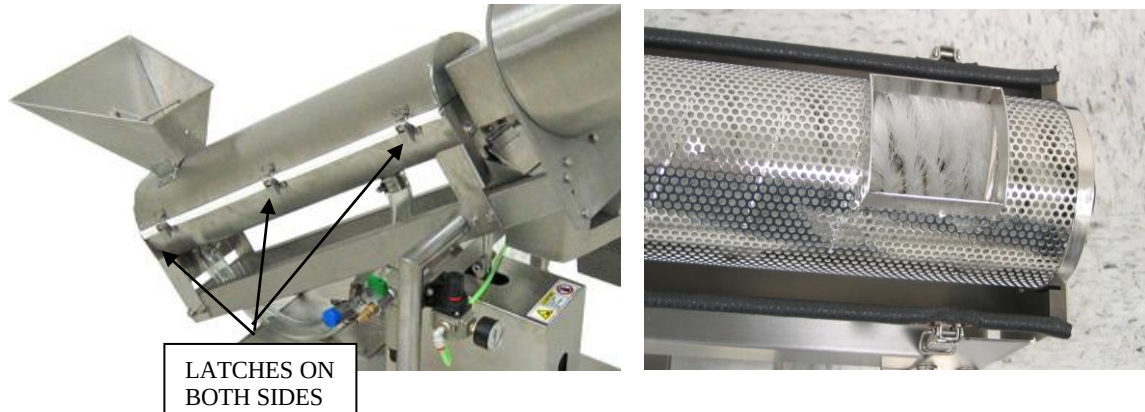
7. Remove deflector (41)

(Re-assembly): Reverse the instructions above.

AIR FLOW BRUSH

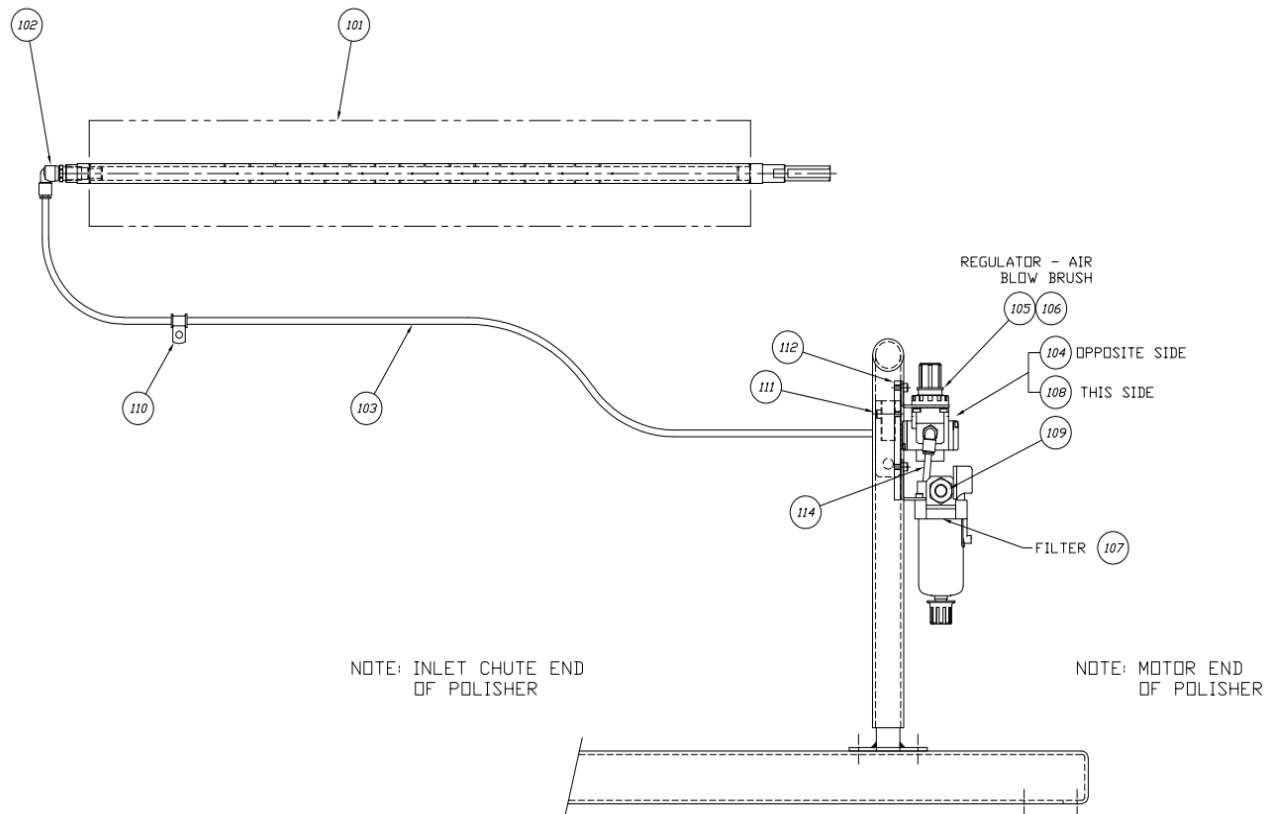
NOTE: DO NOT OPERATE with controller settings above 41 Hz (1200 rpm) with Air Flow Brush Installed.

1. Push the Red Button on the controller to “Stop”, move the ON/OFF switch to the “OFF” position and unplug the CP-4200. Remove the top cylinder half by opening the six (6) cam latches. Inspect and clean as necessary.



2. Remove the shaft coupling guard and loosen the allen set screws that hold the shaft coupling on the brush shaft only (the motor coupling does not have to be loosened). Disconnect the coupling and lift out the brush and polisher tube assembly from the lower tube ends. Slide the brush out of the polisher tube and inspect for strangers.
3. Wash the Air Flow Shaft Brush prior to install if desired. (The brush may be washed in any mild detergent or dish liquid. Washing with solvents is not recommended. If the brush is washed, dry it on the cradles provided to prevent distortion of the bristles.) Remove bearings from both ends of the old brush (keep - to be used on new air flow shaft).
4. Inspect the base cylinder half for strangers.
5. Disconnect the vacuum lines from the base cylinder half. Inspect the lines for strangers and reattach.
6. To reinstall, install the bearing “N” on the new brush shaft (leave bearing loose until mounted in Tube End, Lower (Det#19) & install bearing “C” on the brush (ref: pg. 8). Reinstall the coupling onto the shaft – leave loose at this time). Install the Rotary Male Elbow (Det#102) in the end of the shaft. (Do Not Over-tighten). Slide the air flow brush into the polisher tube assembly. Set the bearings into the “tube end, lower mount” (Det#19) and the “lower tube end, w/cut out” (Det#2) on the base cylinder half. Make sure that the bearings are securely seated in the lower mounts. Set the brush and polisher tube assembly (carries the (2) “tube End Cap, Top (Det#5) Bearing “N” needs to be tightened with set screw onto the flat on the brush shaft) - Reconnect the coupling making sure that the coupling is fully engaged and that the allen set screws are tight. Double-check the tightness of the set- screws in the couplings. Reattach the coupling cover.
7. Reattach the top cylinder half making sure that the bearings are fully seated in their blocks. Secure all of the cam latches.
8. Inspect all of the external surfaces of the polisher, including the base, for strangers: especially in the corners formed by the frame weldment and the end brackets.

AIR FLOW BRUSH Drawing and Part List



Item#	Description	Qty	Part #
101	Brush Shaft, Air Blowing	1	CP0093
102	Rotary Male Elbow Air Fitting	1	CP0098
103	Blue Tubing 6MM OD X 3 Feet Long	1	NA
104	Male Elbow	1	NA
105	Air Regulator	1	CB2008
106	Air Gauge	1	NA
107	Filter	1	CB2002
108	Fitting 1/4 Tube X 14 NPT X 90°	1	CB2005
109	Ball Valve	1	NA
110	Cable Clamp	1	NA
111	10-24 X 3/4 LG BHCS	1	NA
112	Adapter Plate	1	CF0095
114	1/4 od X 16" LG. - Yellow Tubing	1	CF2779
115	10-24X 3/8 LG. SHCS	1	NA

Preventive Maintenance

STI recommends that the CP-4200 Polisher have a three month PM for the following:

- **Check the wires and plug for wear and tear. Repair as needed**
- **Check all air lines/vacuum hose for leaks, Replace as needed**
- **Check the condition of the two bearings p/n's CP0100 and CP0101. These bearings are sealed bearing. If the bearings are found to be in poor working condition, replace them, do not try to grease them**
- **Check the condition of the Spider Coupling p/n CP0055. Replace as needed.**